MEDICOOL

Portable Smart Thermoelectric Medicine Storage System

Concept	A lightweight, rechargeable portable storage device for temperature-sensitive medicines.
Core Technology	Thermoelectric (Peltier) temperature control with insulation, sensing and feedback control.
Primary Use	Portable storage of medicines such as insulin when conventional refrigeration is unavailable.

1. Title of the Project / Product

MediCool – Portable Smart Thermoelectric Medicine Storage System

A lightweight, rechargeable and portable temperature-controlled storage device designed for carrying temperature-sensitive medicines such as insulin during travel and situations where conventional refrigeration is unavailable.

2. Problem Statement

People who depend on temperature-sensitive medicines such as insulin face difficulties maintaining the required storage conditions during travel, outdoor activities, power outages and in locations without reliable access to refrigeration, which can compromise medicine quality and usability.

3. Solution Proposed

MediCool is a compact, rechargeable portable medicine-storage system that maintains a controlled temperature inside an insulated chamber using thermoelectric (Peltier) cooling technology.

The device will consist of an insulated storage chamber, a Peltier thermoelectric module, heatsink and fan assembly, temperature sensor, microcontroller-based control system, rechargeable battery and user interface. The Peltier module transfers heat from the medicine chamber to the external heatsink. The hot side is cooled using the heatsink and fan while the cold side is coupled to the insulated chamber.

The temperature sensor continuously monitors the chamber. A microcontroller regulates the Peltier system according to feedback from the sensor, helping maintain the target range and reduce unnecessary power consumption.

Key Features

• Lightweight and compact design

• Rechargeable battery operation

• Thermoelectric / Peltier cooling

• Real-time temperature monitoring

• Automatic temperature control

• Insulated medicine compartment

• High/low temperature warning

• Battery-level monitoring

• USB-C charging

• Optional ESP32-based Bluetooth/mobile monitoring

Basic Working Mechanism

Rechargeable Battery	Power Management → Peltier Module
Temperature Sensor	Microcontroller → Peltier Control
Peltier Cold Side	Insulated Medicine Chamber → Controlled Environment

4. Implementation Plan

Target Beneficiaries

People carrying temperature-sensitive medicines; insulin-dependent users; travellers; outdoor workers; patients travelling long distances; people in areas with unreliable electricity; and emergency or disaster-response situations where refrigeration is not readily available.

Implementation Approach

Phase 1 – Requirement Analysis	Determine required temperature range, storage capacity, portability and expected operating duration.
Phase 2 – Prototype Development	Build an insulated chamber and integrate the Peltier module, heatsink, temperature sensor and rechargeable battery.
Phase 3 – Control System	Implement microcontroller-based feedback control for temperature monitoring and regulation.
Phase 4 – Testing	Evaluate temperature stability, cooling performance, battery runtime, weight, heat dissipation and recovery after opening.
Phase 5 – Product Development	Improve enclosure, insulation, thermal management, power management and user interface.

Expected Outcomes: A functional prototype capable of maintaining a controlled environment for temperature-sensitive medicines while prioritizing portability and ease of use.

Deliverables: Functional prototype, temperature-control system, rechargeable power system, insulated medicine compartment, monitoring interface, testing report and final product design.

5. Safety and Efficiency Analysis

Safety

• Temperature protection: Continuous monitoring with warnings when the chamber leaves the configured range.

• Over-temperature protection: Monitoring and thermal management of the Peltier hot side.

• Battery protection: Battery Management System (BMS) for overcharge, over-discharge and excessive-current protection.

• Electrical safety: Proper insulation, fusing/current protection and enclosed electrical connections.

• Medicine protection: Control logic designed to avoid excessive cooling or unintended freezing where freezing would be harmful.

Efficiency

Thermal insulation will reduce heat entering the storage chamber, reducing the cooling load. Feedback control will operate the cooling system according to temperature rather than continuously. An efficient heatsink will improve heat rejection from the Peltier hot side, while battery monitoring will help the user manage remaining operating time.

6. Scalability and Future Development

Smartphone connectivity	ESP32-based Bluetooth monitoring of temperature and battery status.
Remote alerts	Notifications when temperature exceeds the configured range or battery becomes critically low.
Data logging	Recording temperature history to help verify storage conditions during transportation.
Improved battery technology	Higher-energy-density batteries to increase runtime while keeping the device compact.
Solar charging	Solar-assisted charging for remote and extended outdoor use.
Multiple medicine compartments	Separate controlled compartments for different storage requirements.
Different form factors	Adaptation into travel cases, backpack-integrated systems and larger medical transport containers.

7. Innovation Advantage

Existing approaches for temperature-sensitive medicine transportation commonly use ice packs, gel packs, insulated bags or conventional refrigeration. These approaches can provide limited cooling duration or require access to a freezer/refrigerator for preparation. MediCool introduces active thermoelectric temperature control in a rechargeable portable form.

Existing Approach	MediCool
Ice / gel packs	Active thermoelectric cooling
Limited cooling duration	Battery-powered active operation
Temperature changes over time	Automatic feedback-based control
Requires pre-cooling / freezing	No ice required
Limited monitoring	Real-time temperature monitoring
Passive insulation	Insulation + active cooling
Basic storage	Smart monitoring and alerts

Core innovation: Combining portability, active thermoelectric temperature control, rechargeable power, insulation and smart monitoring in a compact medicine-storage platform.

8. Impact & User Journey

The proposed user journey highlights the change from dependence on passive cooling or nearby refrigeration to an actively controlled portable storage solution.

BEFORE	TRANSITION	AFTER
Medicine needs controlled conditions	User prepares MediCool	Medicine placed inside insulated chamber
Travelling / power outage	Device switched on	Peltier system actively controls temperature
Search for refrigerator / ice	Sensor measures chamber temperature	Temperature displayed continuously
Carry ice packs / bulky cooling bags	Controller regulates cooling	Portable rechargeable operation
Constant concern about temperature	Alerts provide feedback	Greater convenience and confidence

Proposed visual storyboard: Medicine required → travel / no reliable refrigeration → place medicine in MediCool → automatic temperature monitoring and thermoelectric cooling → portable controlled storage.

Technical Concept – Prototype Architecture

The prototype will use a closed-loop temperature-control architecture. The insulated chamber minimizes heat transfer from the environment. A temperature sensor provides feedback to the microcontroller, which controls the thermoelectric cooling stage. Heat removed from the chamber must be rejected to the ambient environment through the heatsink and fan.

Component	Purpose
Insulated chamber	Minimizes external heat entering the medicine compartment.
Peltier module	Transfers heat from the chamber to the hot side using electrical power.
Heatsink	Rejects heat from the Peltier hot side to the environment.
Temperature sensor	Measures the storage-compartment temperature.
Microcontroller	Runs feedback control, display and alarm logic.
Battery + BMS	Provides portable power and battery protection.
Display / indicators	Shows temperature, operating state and battery status.
Optional ESP32	Enables Bluetooth-based monitoring and future data logging.

Prototype Development Priorities

• Determine the medicine-specific target temperature range from authoritative product/storage requirements.

• Select Peltier capacity based on calculated heat load and expected ambient conditions.

• Select insulation thickness and material to minimize thermal leakage.

• Size the battery for the intended operating duration while considering Peltier power consumption.

• Design reliable heat rejection on the hot side.

• Validate temperature stability, thermal recovery and battery runtime through controlled testing.

Engineering caution: A prototype should not be used to store or administer medicine until its temperature performance, electrical safety, battery safety and reliability have been properly validated. The design must be adapted to the exact storage requirements of the intended medicine.